GGG-P-471E
April 6, 1976
SUPERSEDING
Int. Fed. Spec. GGG-P-00471d (GSA-FSS)
December 3, 1965 and
Fed. Spec. GGG-P-471c
January 8, 1960

FEDERAL SPECIFICATION

PLIERS; PLIERS, SLIP JOINT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers pliers used for gripping operations in assembly and repair work. Some of the pliers are also used for cutting operations.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the titles of the specifications, or which are commercially available, but include only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes and styles. Pliers shall be of the following types, classes, and styles as specified (see 6.1).

- Type I - Battery (without cutter).
- Type II - Slip joint.
  - Class 1 - Angle nose, without cutter.
    - Style A - Multiple tongue and grooves.
  - Class 2 - Straight nose, combination jaw, with cutter.
    - Style A - Regular.
    - Style B - Thin.
  - Class 3 - Bent nose, combination, with cutter.
    - Style A - Regular.
    - Style B - Thin.
- Type III - Combination jaw, lap joint, flat nose (with cutters).
- Type IV - (see 6.2).
- Type V - Flat nose.
  - Class 1 - Narrow nose, without cutter.
  - Class 2 - Medium nose, without cutter.
  - Class 3 - Wide nose.
    - Style A - Without cutter.
    - Style B - With cutter.
- Type VI - Gas, without cutter.
- Type VII - Glass.
- Type IX - Lineman's.
  - Class 1 - With side cutter.
    - Style A - Plain handles.
    - Style B - Insulated handles.
  - Class 2 - With side cutter and wire skinner.
- Type X - Round nose, round jaw (without cutter).
- Type XI - Round nose (chain).
  - Class 1 - Long nose.
    - Style A - With cutter.
    - Style B - Without cutter.

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Class 2 - Short nose.
Style A - With cutter.
Style B - Without cutter.
Type XII - Round nose (needle).
Class 1 - Curved nose (without cutter).
Class 2 - Straight nose.
Style A - Without cutter.
Style B - With cutter.
Type XIII - (See 6.2).
Type XIV - Parallel-action jaw.
Class 1 - With side cutter.
Class 2 - Without cutter.
Type XV - Duckbill, without cutter.
Class 1 - Long nose.
Class 2 - Short nose.
Type XVI - Fence, single head.
Type XVII - (see 6.2).
Type XVIII - Heat-coil.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-P-121 - Paper, Bond and Writing, White and Colored.
PPP-P-40 - Packaging and Packing of Hand Tools.

Federal Standard:

Fed. Std. No. 358 - Sampling Procedures.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests For.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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3. *REQUIREMENTS*

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of pliers which are otherwise in accordance with this specification.

3.2 Material. Pliers shall be forged from steel of such quality that when properly heat treated, will comply with the applicable requirements hereinafter specified except that steel for type XIV pliers may be fabricated from formed or machined steel capable of complying with the requirements specified herein. Joint rivets and bolts may be of other than steel if such material meets all other requirements of this specification.

3.3 Design. The pliers shall be similar to the figure to which reference is made. Pliers shall be properly proportioned in all parts so as to be strong, durable, and easy to operate. The hand span dimension "E" (shown on most figures) and the B dimension on figures 23 and 24 shall be measured with the plier jaws in the closed position.

3.3.1 Cutting.

3.3.1.1 Wire cutting. The type III, type V, class 3, style B, type IX, type XI, classes 1 and 2, style A, type XII, class 2, style B, type XIV, class 1 and type XVI pliers shall cut wire without any assistance, such as bending, pulling or twisting of the wire, and there shall be no visible deformation or damage to the cutting edges of the cutter when tested as specified in 4.3.3.1.1.

3.3.1.2 Paper cutting. Following completion of the wire cutting test of 4.3.3.1.1, the type V, class 3, style B, type IX, type XI, classes 1 and 2, style A, type XII, class 2, style B, and type XIV, class 1 pliers shall completely and cleanly cut bond paper without exceeding the load specified herein (see 4.3.3.4).

3.3.1.3 Nail cutting. The type II, classes 2 and 3 pliers shall be capable of withstanding the nail cutting test of 4.3.3.3 without any visible deformation or damage to the cutter as a result of the test,

3.3.1.4 Lamp cord cutting. The type IX pliers shall be capable of withstanding the lamp cord cutting test of 4.3.3.2.

3.3.2 Wire bending. The type V, class 2, type V, class 3, styles A and B, type X, type XI, type XII, class 2, styles A and B, type XIV, and type XV pliers shall show no cracks, deformation, looseness, or variation in alignment of the jaws or handles as a result of the test specified in 4.3.4.

3.4 Handles. Handles of pliers shall be so shaped as to afford a comfortable grip and shall be free from rough edges and sharp corners. Inner handle surfaces adjacent to the joint shall be well-rounded or beveled so as to prevent injury of the hands. Except for type XIV pliers, the handles shall have a Rockwell "C" hardness of from 35 to 50 when tested as specified in 4.3.1. Ends of handles of pliers shall not touch when the jaws are in the closed position, except those with slip joints in the fully extended position.

3.4.1 Allowable deflection and set. The deflection under load and permanent set of the handles for all type pliers except type XIV, class 2 shall not exceed the values specified herein when subjected to the applicable load test specified in 4.3.2.

3.4.2 Insulated handles. Type IX, class 1, style B plier handles shall be insulated to meet the dielectric strength test in 4.3.6. All other varieties of pliers except type IX, class 1, style A may be furnished with covered handles at the option of the contracting officer (see 6.1), or the supplier provided they meet the dielectric strength test in 4.3.6. One eighth inch additional tolerance will be allowed for insulation on applicable dimensions.

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3.5 Joints. Pliers shall have either lap or box joints. There shall be no excessive sidewise movement, play or other indication of looseness of the two halves of the tool when the pliers are open or closed. The Rockwell "C" value of the joint rivet or bolt shall be not less than 28 nor more than 50 for all pliers with the exception of types I and II, classes 1, 2, and 3, and type XIV. Where joint rivet on bolt receives a casehardening treatment in addition to the through hardening, a maximum hardness of 60 on the Rockwell "C" scale will be permitted. The bolts, pins, or rivets, when furnished on types I and II, classes 1, 2, and 3 pliers shall show a hardness value of not less than 35 nor more than 45 on the Rockwell "C" scale or be casehardened to a minimum depth of 0.010 inch and show a hardness value of not less than 78 nor more than 90 on the Rockwell 15 N scale (superficial). The center screw of type XIV, class 1 pliers shall be casehardened to a depth of not less than 0.003 inch (file hard). The center screw or joint rivet of type XIV, class 2 pliers may be either hardened or unhardened at the option of the manufacturer.

3.6 Jaws.

3.6.1 Jaw openings. The ends of jaws shall open to the respective minimum distance specified herein for the individual type, class, style, and size pliers, through which distance the tool shall operate in a smooth and uniform manner. It shall be possible to open the handles through the minimum jaw opening specified herein by the application of a force to the handles of the magnitude specified herein and in the manner specified in 4.3.5. Beyond the minimum jaw opening distance specified herein, the jaws may open at increased loads until the positive stop of the tool is engaged.

3.6.2 Jaw hardness. Except as otherwise specified in this paragraph, jaws shall show a hardness value of not less than 45 and not more than 60 on the Rockwell "C" scale when tested as specified in 4.3.1. Side-cutting pliers shall have a Rockwell "C" hardness between 55 and 65 adjacent to the cutting edge. Type XIV plier jaws and cutter blades shall be either through hardened or casehardened to the hardness specified in this paragraph.

3.7 Scoring. Scored surfaces, where specified for gripping jaws, shall have sharp projections which shall not be retouched or refinished by filing or grinding.

3.8 Finishes.

3.8.1 Surfaces. All surfaces shall be free from pits, nodules, burrs, cracks, and other defects which may adversely affect the performance of the tool. Surfaces usually ground or otherwise finished and provided with one of the coatings specified in 3.8.2 shall have a maximum surface roughness of 63 microinches (arithmetical average).

3.8.2 Coatings. Unless otherwise specified (see 6.1), types I and II pliers shall have polished heads and chromium plated all over; types III, V, VI, VII, IX, X, XI, XII, XV, and XVIII shall have polished heads and chemically produced oxide or phosphate coated handles with a supplementary oil treatment. Types XIV and XVI shall have a chemically produced oxide or phosphate coating with a supplementary oil treatment, or a corrosion-resisting treatment in accordance with the manufacturer's standard commercial practice.

3.9 Marking. The pliers shall be marked in a plain and permanent manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.10 Type I. Type I pliers shall be suitable for handling storage-battery terminals and connecting lugs. The gripping surfaces shall be deeply scored the full length of the jaws with sharp milled parallel crosswise grooves. The jaws shall open and close in a uniform manner from their minimum opening position to a parallel position. The jaws shall be offset 15 to 40 degrees as measured from the center line of the jaws and the center line of the handles. With the pliers parallel to the longitudinal axis and set in a gripping position on a 5/8-inch-diameter rod, in a manner simulating the gripping of a storage-battery terminal for lifting purposes, the maximum distance between the outsides of the handles at their point of widest separation shall not exceed 4-1/2 inches and the gripping portion shall not exceed 4 inches. The pliers shall conform to the requirements shown in table I and shall be similar to figure 1. (Pliers with a two-position slip joint which otherwise meet the foregoing requirements are a suitable alternate.)

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TABLE I. Type I

Nominal Size	A ± 1/2	B ± 1/4	C ± 1/16	D ± 1/8	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inch	Inch
7 ----	7-1/2	1-3/8	7/32	3/8	5/8

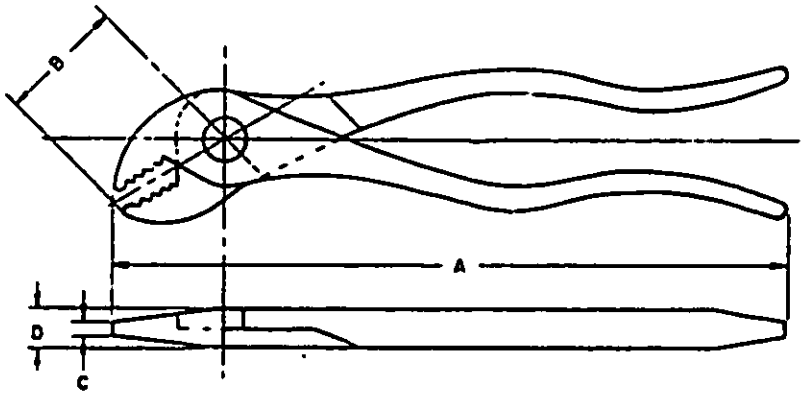


Figure 1. Type I

3.11 Type II.

3.11.1 Class 1. Class 1 pliers shall have jaws offset 37 to 56 degrees as measured between the centerline of the jaws and the centerline of the handles. The gripping surface shall have sharp, crosswise milled grooves the full length of the jaws. With the jaws parallel in any of the operating positions, the maximum distance between the outsides of the handles at their point of widest separation shall not exceed 2-1/2 inches. With the jaws parallel and in the outermost operating position, the maximum work capacity of the jaws shall be not less than the minimum work capacity of the pliers as specified in table II. Handles shall not contact each other when the jaws are parallel in any of the operating positions and shall be of approximately equal length when the pliers are fixed in the central operating position.

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3.11.1.1 Style A. Style A pliers shall be similar to figure 2, except size 10A which shall be similar to figure 3, and conform to the requirements shown in table I for the size specified (see 6.1). The slip joint shall consist of a number of curved grooves on one side of the plier half, and a curved tongue on one side of the other plier half. The two plier halves shall be held together by a bolt and nut. Changes in jaw position shall be made by sliding the bolt forward or backward in a centrally located slot in the grooved plier half until the tongued side is aligned with, and engaged in the desired groove. The contacting sides of the tongue and grooves shall be essentially perpendicular to their base, or undercut to prevent a tendency of the pliers to separate under pressure exerted on the handles when being used. The tongue and grooves shall be suitably finished to provide a smooth engagement in each of the operating positions, and the bolt and nut shall be assembled so that each plier half slides freely without binding. The nut shall fit the bolt with a prevailing torque, or be staked, so as to prevent loosening under service conditions.

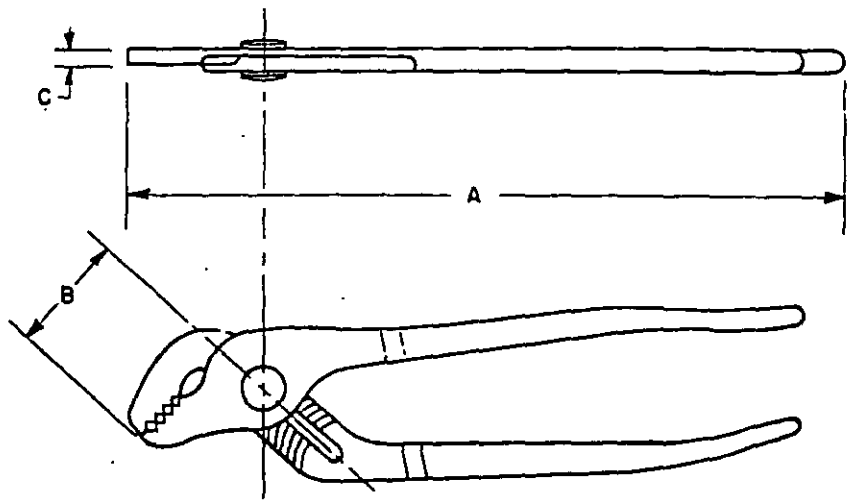


Figure 2. Type II, class 1, style A (straight jaw).
Angle nose, without cutter, multiple grooves and tongue.

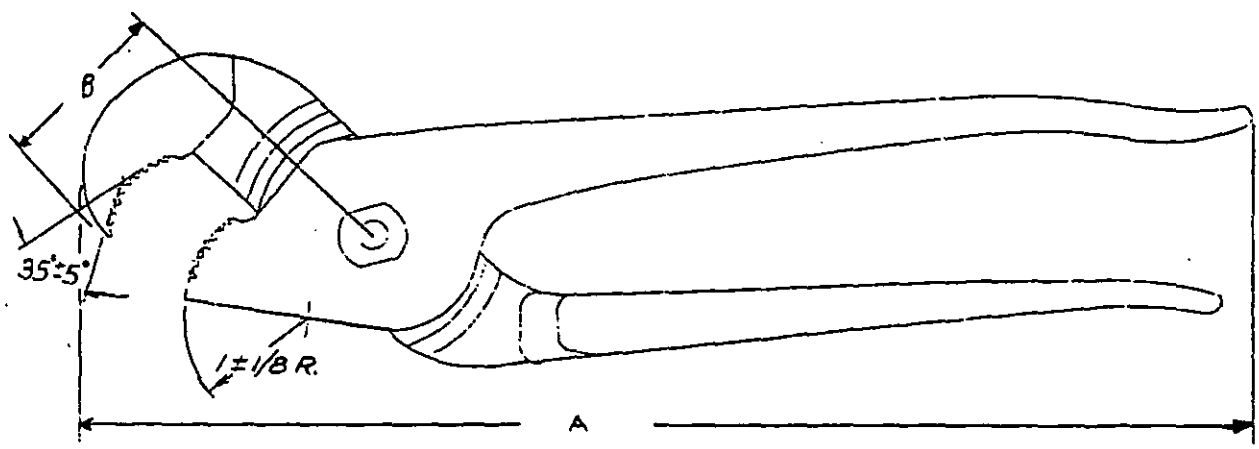


Figure 3. Type II, class 1, style A (parrot nose).
Angle nose, without cutter, multiple grooves and tongue.

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TABLE II. Type II, class 1, style A.
Angle nose, without cutter, multiple grooves and tongue

Nominal size	A ±1	B +1/4 -1/6	Jaw[1] thickness C +1/8 -1/32	Work capacity minimum	Number of adjustment positions minimum
Inches	Inches	Inches	Inch	Inches	
5----------	5	3/4	1/8	7/16	3
8----------	8	1-3/8	1/4	11/16	4
10----------	10	1-5/8	9/32	1-3/8	4
10A---------- 2/	10	1-3/8	7/16	1-1/8	4
12----------	12	2	7/16	2-1/2	6
16----------	16	2-7/8	1/2	4	10

[1]Measured 1/4 inch from tip of jaw to provide for acceptance of tapered jaws.
[2]Shall be of "parrot nose" design.

3.11.2 Class 2. Class 2 pliers shall have a straight nose and shall consist of combination jaws, slip joint, and a wire cutter of either the side or throat type. The combination jaws shall have straight, scored gripping surfaces at the outer end of the jaws, suitable for gripping flat surfaces, and curved gripping jaws with pointed teeth at the center of the jaws, suitable for gripping round surfaces. The slip joint shall allow for ready use of the jaw openings in normal or extended positions, by slipping the double-hole side of the pliers on a bolt which is stationary on the single-hole side. The holes of both the double-hole side and single-hole side shall have sharp and accurately sized openings for the bolt. The bolt shall be of the round-head type, threaded and fitted with a compact nut arranged to permit adjustment of the two halves of the pliers. The nut shall be secured so as to prevent its removal under ordinary service conditions. The pliers shall conform to the requirements of 3.3.1.3.

3.11.2.1 Style A. Style A pliers shall conform to the requirements shown in table III for the size specified (see 6.1), and shall be similar to figure 4.

TABLE III. Type II, class 2, style A

Nominal size	A ±1	B +1/4 -1/2	C +1/8 -3/32	E ±3/8
Inches	Inches	Inches	Inch	Inches
6----------	6	1-3/4	3/8	1-3/4
8----------	8	1-7/8	3/8	1-7/8
10----------	10	2-3/8	17/32	2-1/8

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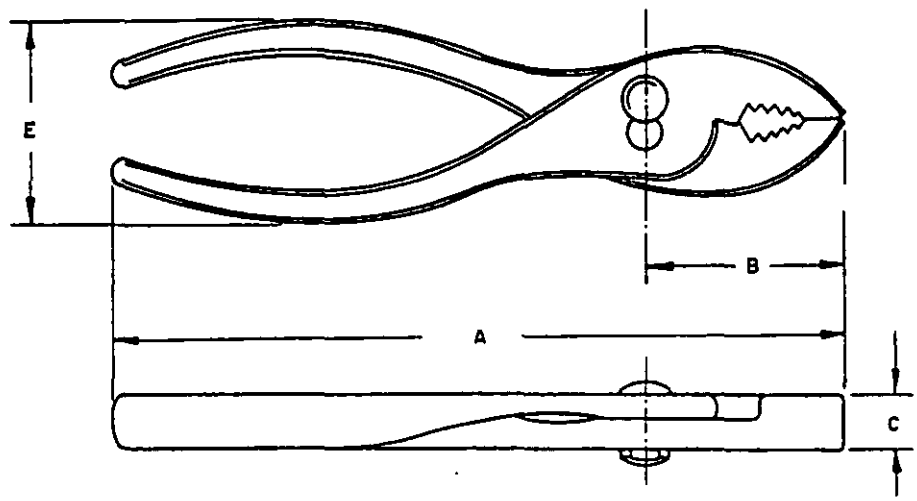


Figure 4. Type II, class 2, style A.
Straight nose, combination jaw, with cutter, regular

3.11.2.2 Style B. Style B pliers shall conform to the requirements shown in table IV for the size specified (see 6.1), and shall be similar to figure 5.

TABLE IV. Type II, class 2, style B

Nominal size	A ±1/2	B ±1/4	C ±3/64	E ±3/8
Inches	Inches	Inches	Inch	Inches
6----------	6-1/4	1-3/4	3/16	1-3/4

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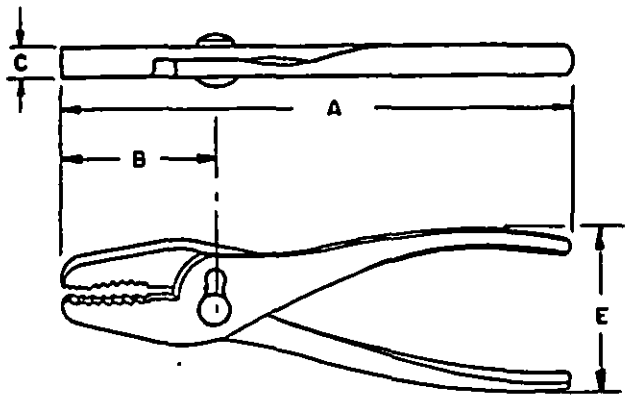


Figure 5. Type II, class 2, style B.
Straight nose, combination jaw, with cutter, thin.

3.11.3 Class 3. Class 3 pliers shall have a nose bent to an angle approximately 25 to 30 degrees. The pliers shall have a cutter, shall conform to 3.3.1.3 and the requirements shown in table V for the size specified (see 6.1).

TABLE V. Type II, class 3, styles A and B

Nominal size	A ±1/2	B ±5/16	C		E ±3/8
			Style A ±1/8	Style B ±3/64	
Inches	Inches	Inches	Inch	Inch	Inches
6----------	6-1/4	1-3/4		3/16	1-3/4
8----------	8-1/4	1-7/8	3/8		1-7/8

3.11.3.1 Style A. Style A pliers shall conform to the requirements of 3.11.3 for the size specified (see 6.1), and shall be similar to figure 6.

3.11.3.2 Style B. Style B pliers shall conform to the requirements of 3.11.3 for the size specified (see 6.1), and shall be similar to figure 6.

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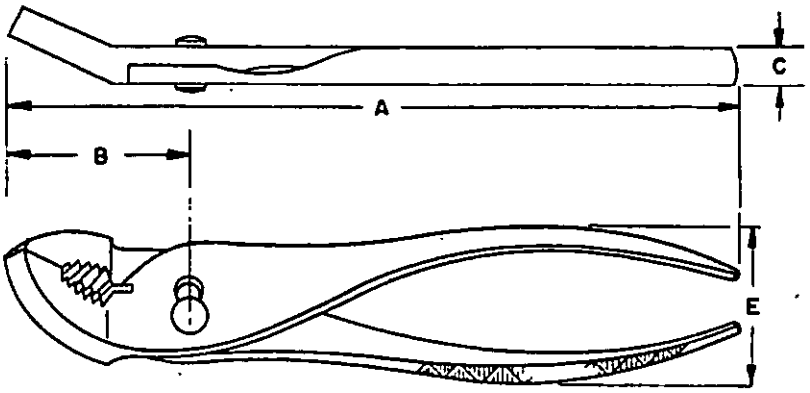


Figure 6. Type II, class 3, styles A and B

3.12 Type III. Type III pliers shall have a lap joint, combination jaws and at least three wire cutters of the button type (see 3.3.1.1). The pliers shall be of flat-nose construction. The combination jaws shall have straight or diamond-shaped scored gripping surfaces at the outer end suitable for gripping flat surfaces, and curved gripping surfaces with pointed teeth at the center, suitable for gripping a 1/4 inch diameter rod securely when the outermost end of the jaws is opened not more than 3/32 inch. The portion of the handles adjoining the joint may be with or without scored faces for gripping wire in stretching. The pliers shall conform to the requirements shown in table VI for the size specified (see 6.1), and shall be similar to figure 7.

TABLE VI. Type III, pliers, combination utility, flat-nose, with wire-cutter

Nominal size	A ±1/2	B ±1/4	C ±1/8	D ±3/16	E ±1/4	F ±1/16	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inches	Inches	Inch	Inch
8----------	8	1-3/8	5/8	1-3/16	1-7/8	1/4	7/8

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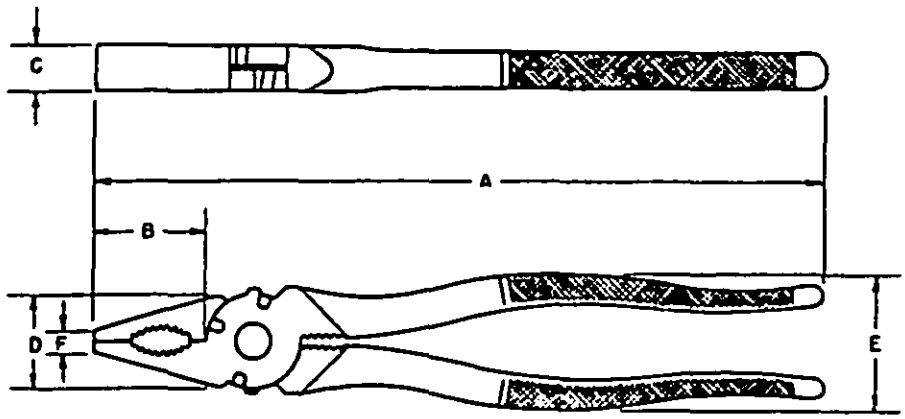


Figure 7. Type III

3.13 Type V. The jaws of type V pliers shall taper as specified for each class. The jaws shall contact each other at the outermost end when the pliers are in a closed position.

3.13.1 Class 1. Class 1 pliers shall have scored jaws and taper uniformly in thickness and in width from near the joint to the outermost end. The pliers shall conform to the requirements shown in table VII and be similar to figure 8.

TABLE VII. Type V, class 1

Nominal size	A	B	C	D	E	F	G	Jaw opening minimum
Inches	Inches	Inches	Inch	Inch	Inches	Inch	Inch	Inch
5-1/2	5-1/2 ± 1/8	1-1/2 ± 1/16	1/4 ± 1/32	1/2 ±1/32	1-7/8 ± 1/4	0.070 ± 0.032	3/16 ± 1/32	7/8

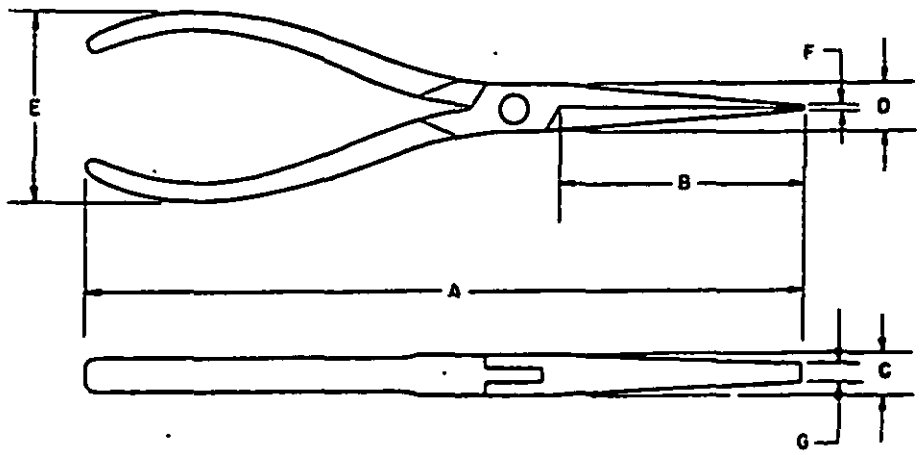


Figure 8. Type V, class 1

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3.13.2 Class 2. The width of the jaws shall be uniform throughout and shall taper in thickness from near the joint to the outermost end. Unless otherwise specified (see 6.1), one-fourth to one-third of the gripping surfaces of the jaws from the outermost end shall be scored to form a diamond point surface. The pliers shall conform to the requirements shown in table VIII for the size specified (see 6.1), and shall be similar to figure 9.

TABLE VIII. Type V, class 2

Nominal size	A ± 1/4	B ± 1/4	C + 1/8 - 1/16	D ± 1/8	E ± 1/4	F + 1/16 - 1/32	G ± 1/16	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inch	Inches	Inch	Inch	Inch
4---------	4-1/4	1	1/4	7/16	1-1/2	1/16	7/32	3/8

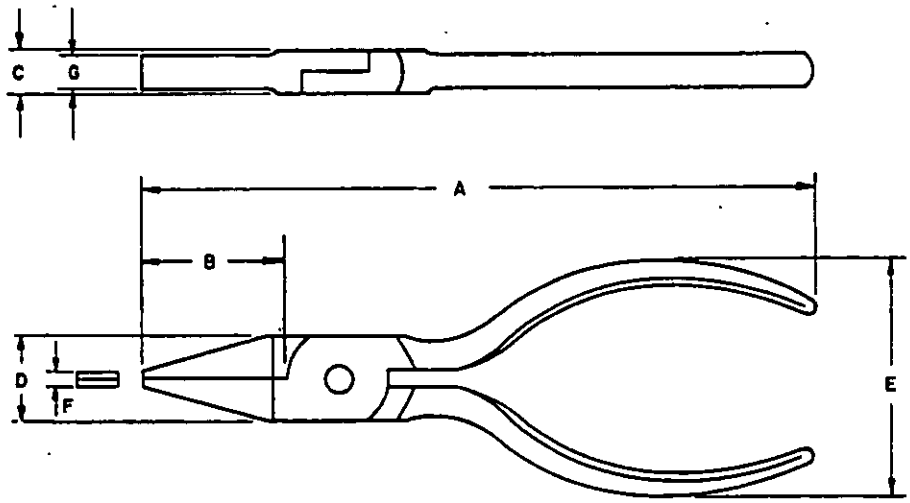


Figure 9. Type V, class 2.

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3.13.3 Class 3. The gripping jaws shall be of uniform width and shall taper in thickness from near the joint to the outermost end. Unless otherwise specified (see 6.1), one-fourth to one-third of the gripping surfaces of the jaws from the outermost end shall be scored.

3.13.3.1 Style A. Style A pliers shall conform to the requirements shown in table IX and shall be similar to figure 10.

TABLE IX. Type V, class 3, styles A and B

Nominal size	A ± 1/2	B ± 1/4	C ± 1/16	D ± 1/8	E ± 1/4	F ± 1/32	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inch	Inches	Inch	Inch
4 ------	4-1/4	1	3/8	7/16	1-1/2	1/16	3/8
6 ------	6-1/2	2-1/4	3/8	11/16	1-7/8	1/8	3/4

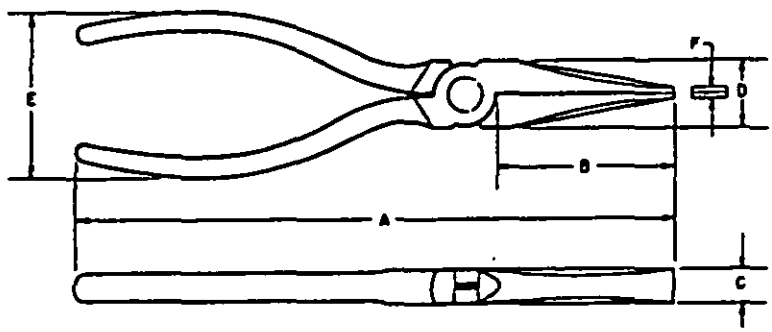


Figure 10. Type V, class 3, styles A and B.

3.13.3.2 Style B. Style B pliers shall be identical to style A pliers, except that the jaws shall have cutting edges on one side near the joint end. The cutting edges shall extend approximately one-fifth the length of the jaws. There shall be a recess in the jaws behind the side cutters to provide satisfactory side clearance.

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3.14 Type VI. The jaws of type VI pliers shall be of uniform width and the outer end or nose shall be rounded. The pliers shall have a central longitudinal grip at the outer end of the jaw, and two different size transverse grips of elliptical shape in the jaw body (see 3.14.1). The gripping surfaces shall have sharp pointed teeth. The front end of the pliers in conjunction with the longitudinal grip shall be suitable for gripping both flat and round objects.

3.14.1 Transverse pipe grips. With the outermost ends of the jaws opened not more than 1/8 inch, the smaller transverse pipe grip shall securely hold an 11/32-inch diameter rod and the larger grip shall securely hold a 7/16-inch-diameter rod, individually. The larger transverse grip shall have a capacity for holding a rod at least 1-1/2 inches in diameter.

3.14.2 Type VI pliers shall conform to the requirements shown in table X and shall be similar to figure 11.

TABLE X. Type VI

Nominal size	A ± 1/2	B ± 1/4	C ± 1/16	D ± 1/8	E ± 3/8	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inches	Inches	Inches
10 --------	10	2-1/8	5/8	1-1/4	1-1/2	1-3/8

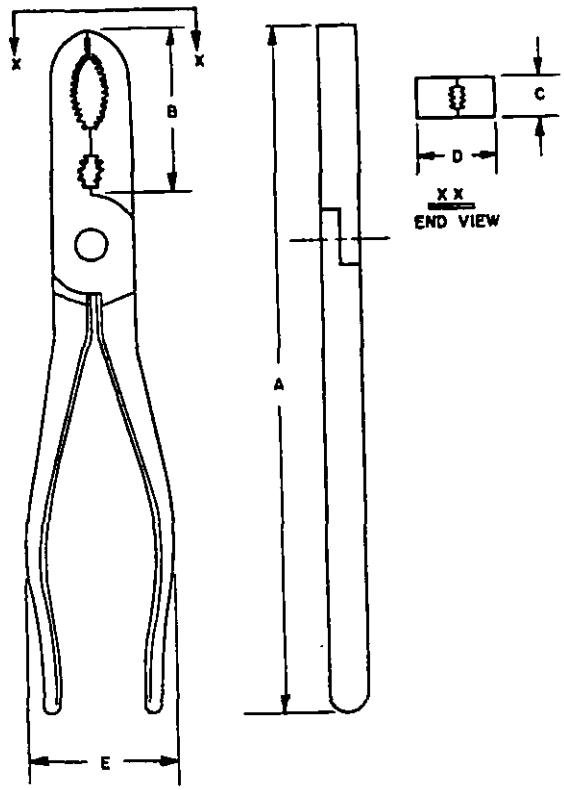


Figure 11. Type VI, Gas, without cutter

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3.15 Type VII. The pliers shall be suitable for firmly grasping plate glass along the full width of the jaw. With the pliers in a close position, the jaws shall contact each other only at their outermost end. With the jaws of the pliers opened so that the jaw surfaces are parallel, the jaw opening shall be not less than 7/16 inch and not more than 1/2 inch. The gripping surfaces of the jaws shall be smooth and without scoring. The pliers shall conform to the requirements shown in table XI and shall be similar to figure 12.

TABLE XI. Type VII

Nominal size	A ± 1/2	B ± 1/4	C ± 1/16	D ± 1/8	E ± 1/4	F ± 1/8	G ± 1/4	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inches	Inches	Inch	Inch	Inches
8 -------	8	1-3/8	1/2	1-1/4	1-7/8	5/16	7/8	1

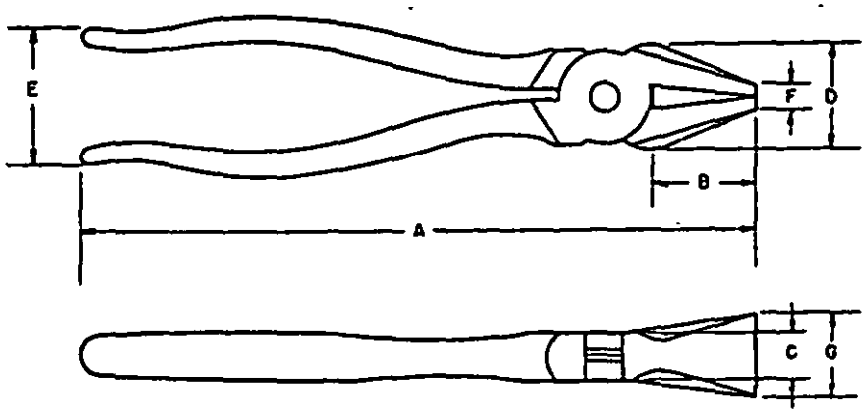


Figure 12. Type VII

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3.16 Type IX.

3.16.1 Class 1. The jaws shall have cutting edges on one side near the joint end. The cutting edges shall extend approximately one-half the length of the jaws. The remaining portion of the jaws shall have straight surfaces, scored with a straight or uniform diamond-shaped pattern. With the pliers in a closed position, the cutting edges shall contact each other throughout the entire length. There shall be a recess in the jaws behind the side cutters to provide satisfactory cutting clearance. The handles at the joint end shall be suitable for crushing wire insulation in accordance with the dimensional requirements shown in figure 13. With the pliers closed, the ends of the jaws shall not touch, but shall leave a clearance which will permit gripping firmly a wire 0.025 inch in diameter using the 6-inch pliers and 0.040 inch in diameter using the 8-inch pliers. The load required to cleanly cut lamp cord shall not exceed the value specified herein, when tested as specified in 4.3.3.2. Class 1 pliers shall conform to the requirements shown in table XII for the size specified (see 6.1), and shall be similar to figure 13.

TABLE XII. Type IX, class 1

Nominal size	A	B	C	D	E1/	F	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inches	Inches	Inch	Inch
6	6-1/2 ± 1/2	1-1/4 ± 1/4	1/2 ± 1/8	7/8 ± 1/8	1-3/4 ± 1/4	7/32 ± 1/16	5/8
8	8-1/2 ± 1/2	1-1/2 ± 1/4	5/8 ± 1/8	1-1/16 ± 1/8	1-7/8 ± 1/4	9/32 ± 1/16	7/8

1/ 1/8-Inch additional tolerance will be allowed for insulation on style B pliers, on applicable dimensions.

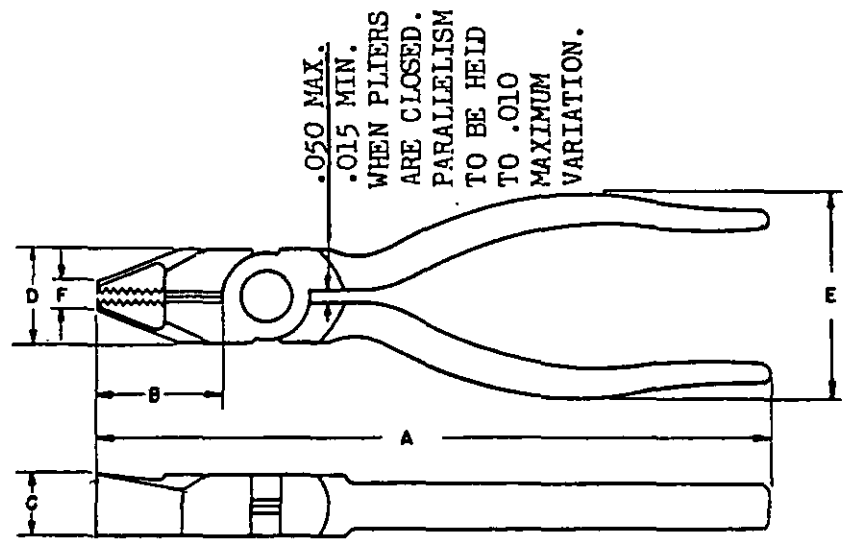


Figure 13. Type IX, class 1

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3.16.1.1 Style A. Style A pliers shall have bare metal handles.

3.16.1.2 Style B. Style B pliers shall be insulated with a durable rubber or plastic coating. The insulation covering shall meet the dielectric strength test specified in 4.3.6.

3.16.2 Class 2. Class 2 pliers shall conform to 3.16.1, except that they shall have two skinning holes drilled in the cutting edges and the pliers shall conform to the requirements shown in table XIII and shall be similar to figure 14.

TABLE XIII. Type IX, class 2

Nominal size	A ± 1/2	B + 3/8 - 1/4	C ± 1/8	D ± 1/8	E ± 1/4	F ± 1/16	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inch	Inches	Inch	Inch
6	6-1/2	1-1/4	1/2	7/8	1-3/4	7/32	5/8
8	8-1/2	1-1/2	5/8	1-1/16	1-7/8	9/32	7/8

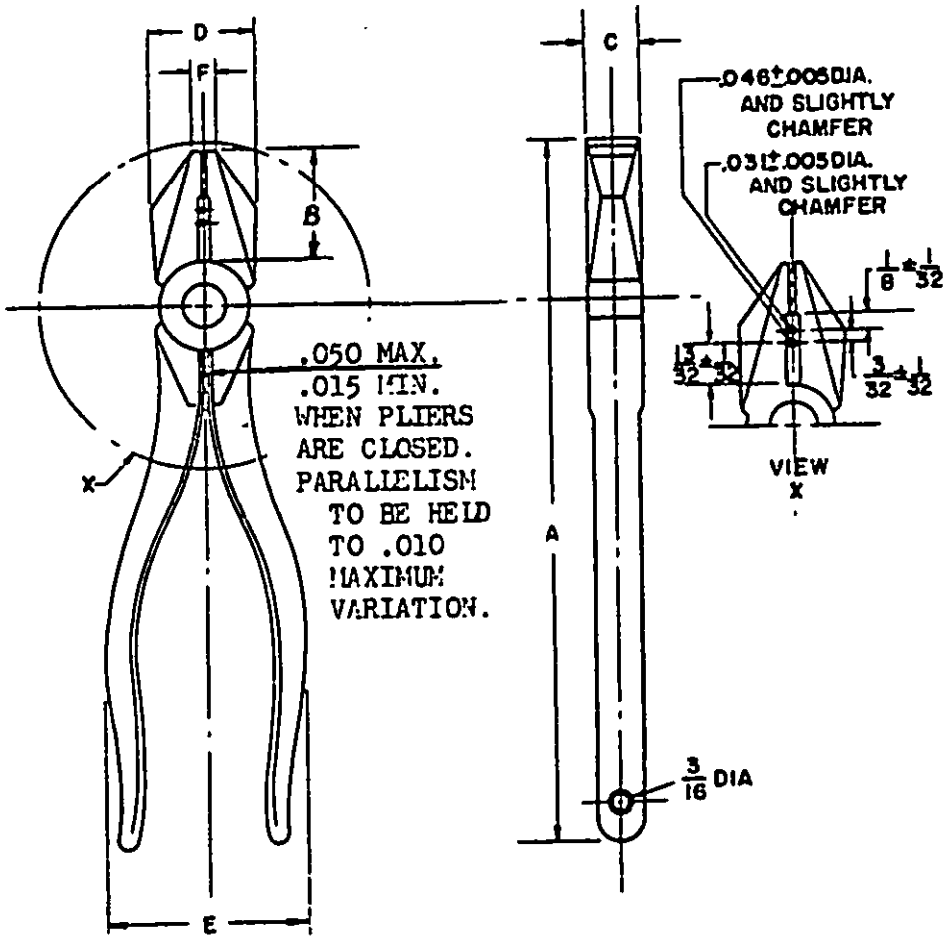


Figure 14. Type IX, class 2

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3.17 Type X. Type X pliers shall be suitable for bending and shaping wire. The jaws shall be round in cross section so that the gripping surfaces are circular. The jaws shall taper uniformly from near the joint to the outermost end. With the pliers in a closed position, the jaws shall contact each other only at the outermost end. With the jaws parallel to each other, the distance between jaws shall be not less than 1/32 inch. The pliers shall conform to the requirements shown in table XIV for the size specified (see 6.1), and shall be similar to figure 15.

TABLE XIV. Type X

Nominal size	A	B	C	D	E	F	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inch	Inches	Inch	Inch
4-1/2	4-1/2 ± 1/4	13/16 ± 1/8	1/4 ± 1/16	7/16 ± 1/16	1-3/4 ± 1/8	3/64 ± 1/32	3/8
6	6-1/4 ± 1/2	1-3/8 ± 1/4	7/16 ± 1/16	5/8 ± 3/16	2 ± 1/4	3/32 ± 1/32	5/8

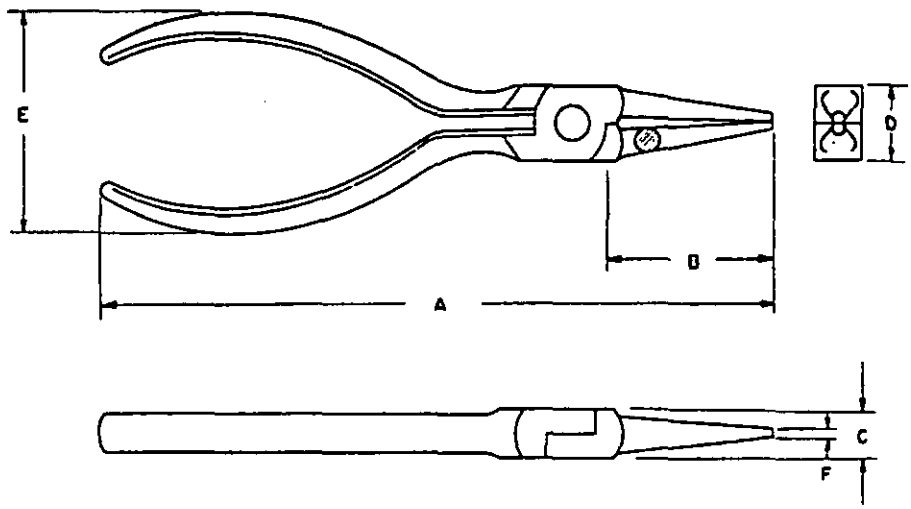


Figure 15. Type X

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3.18 Type XI. The jaws of type XI pliers shall be suitable for forming wire loops, handling small objects, and reaching into small openings. Each jaw shall be of half-round cross section so that the gripping surfaces are planer and straight. The jaws shall taper uniformly from near the joint section to the outermost end. One-fourth to one-third of the gripping surfaces of the jaw from the outermost end shall be scored. The jaws shall contact each other at the outermost end when pliers are in a closed position.

3.18.1 Class 1.

3.18.1.1 Style A. The jaws shall be provided with cutting edges on one side adjacent to the joint and extending one-fourth the length of the jaws. With the pliers in the closed position, the cutting edges shall contact each other throughout the entire length. There shall be a recess behind the cutter to provide clearance for the cutting edges. The pliers shall conform to the requirements shown in table XV for the size specified (see 6.1), and shall be similar to figure 16.

TABLE XV. Type XI, class 1, style A

Nominal size	A ± 3/8	B ± 1/4	C ± 1/16	D ± 1/8	E ± 1/4	G ± 1/32	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inch	Inches	Inch	Inches
6 -----	6-1/2	2	3/8	11/16	1-7/8	1/8	1
7 -----	7-1/2	2-3/4	3/8	3/4	1-7/8	1/8	1-1/8

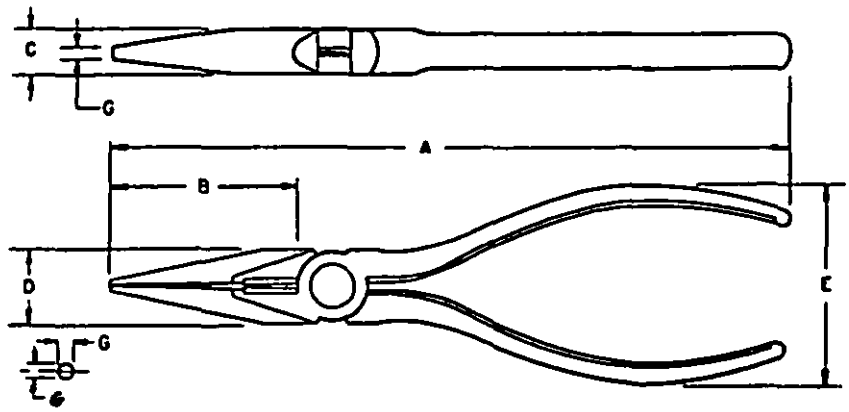


Figure 16. Type XI, class 1, style A

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3.18.1.2 Style B. Style B pliers shall conform to the requirements shown in table XVI for the size specified (see 6.1), and shall be similar to figure 17.

TABLE XVI. Type XI, class 1, style B

Nominal size	A	B	C	D	E	G	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inch	Inches	Inch	Inches
4-1/2	4-5/8 ± 1/8	1-1/8 ± 1/16	1/4 ± 1/32	7/16 ± 1/32	1-5/8 ± 1/8	3/64 ± 1/64	3/4
6	6-1/4 ± 1/4	2 ± 1/4	3/8 ± 1/16	5/8 ± 1/8	1-7/8 ± 1/4	1/16 ± 1/32	1

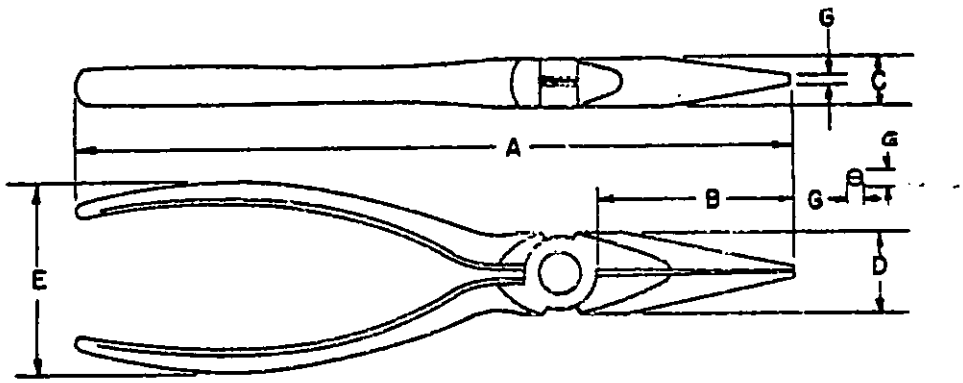


Figure 17. Type XI, class 1, style B

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3.18.2 Class 2. The jaws of class 2 pliers shall be suitable for forming wire loops, handling small objects, and reaching into small openings. Each jaw shall be of half round cross section so that the gripping surfaces are planar and straight. The jaws shall taper uniformly from near the joint section to the outermost end. One-fourth to one-third of the gripping surfaces of the jaws from the outermost end, shall be scored. The jaws shall contact each other at the outermost end when pliers are in the closed position.

3.18.2.1 Style A. The jaws shall be provided with cutting edges on one side adjacent to the joint and extending approximately one-fourth the length of the jaws. There shall be a recess behind the cutter to provide satisfactory clearance for the cutting edges. The pliers shall conform to the requirements shown in table XVII and be similar to figure 18.

TABLE XVII. Type XI, class 2, style A

Nominal size	A ± 1/4	B ± 1/4	C ± 1/16	D ± 1/8	E ± 1/4	G ± 1/32	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inch	Inches	Inch	Inch
5 -----	5-1/4	1-1/4	3/8	1/2	1-3/4	1/16	3/8
6 -----	6-1/4	1-1/2	3/8	5/8	2	1/16	5/8

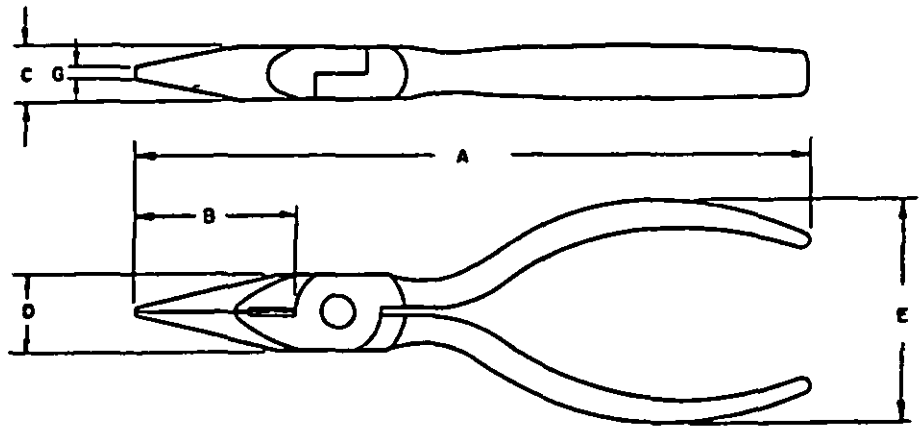


Figure 18. Type XI, class 2, style A

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3.18.2.2 Style B. Style B pliers shall conform to the requirements shown in table XVIII for the size specified (see 6.1), and shall be similar to figure 19, except that the 8-inch size may be similar to figure 20.

Table XVIII. Type XI, class 2, style B

Nominal size	A ± 1/4	B ± 1/4	C ± 1/16	D ± 1/8	E ± 3/8	G ± 1/32	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inch	Inches	Inch	Inch
4 -----	4-3/8	1	1/4	3/8	1-5/8	1/16	3/8
5 -----	5-1/4	1-1/4	3/8	1/2	1-7/8	1/16	1/2
6 -----	6-1/4	1-1/2	3/8	5/8	2	1/16	5/8
8 -----	7-3/4	1-1/2	7/16	9/16	2	1/16	3/4

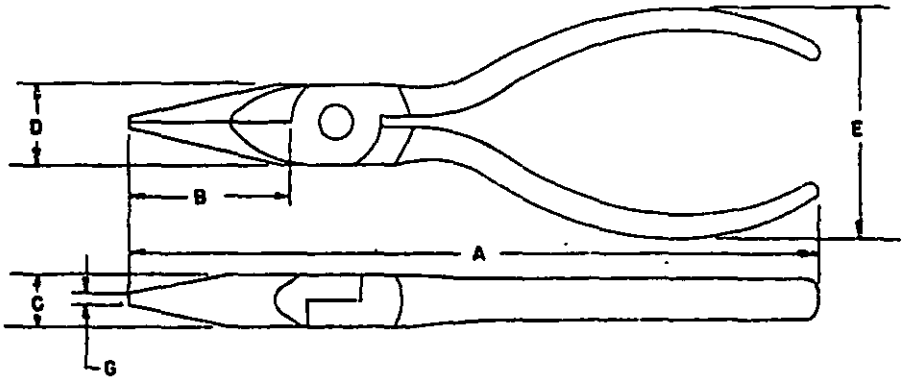


Figure 19. Type XI, class 2, style B

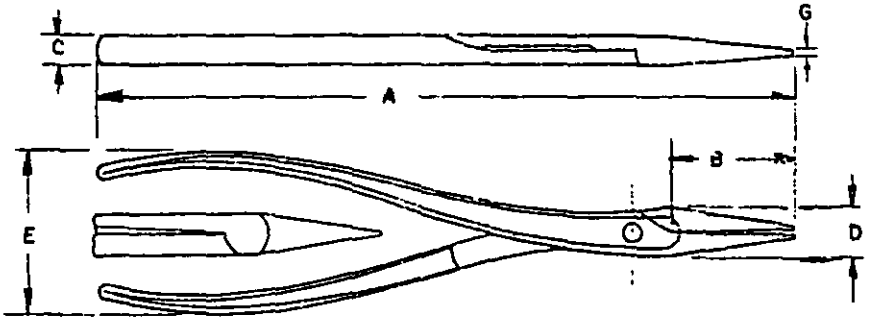


Figure 20. Type XI, class 2, style B (8-inch size), optional design

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3.19 Type XII. The jaws of type XII pliers shall be of half-round cross section so that the gripping surfaces are planar. Unless otherwise specified (see 6.1), one-fourth to one-third of the gripping surface of the jaws from the outermost end shall be scored. The jaws shall contact each other at the outermost end when the pliers are in a closed position.

3.19.1 Class 1. The jaws of class 1 pliers shall be curved and shall taper from near the joint section or the bend to the outermost end. The pliers shall conform to the requirements shown in table XIX and be similar to figure 21.

TABLE XIX. Type XII, class 1

Nominal size	A ± 1/2	B ± 1/4	C ± 1/16	D ± 1/4	E + 1/2 - 1/4	G ± 1/32	H Min.	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inch	Inches	Inch	Inches	Inch
6 -----	6-1/4	1-1/8	3/8	11/16	1-7/8	3/32	1-1/8	1/2

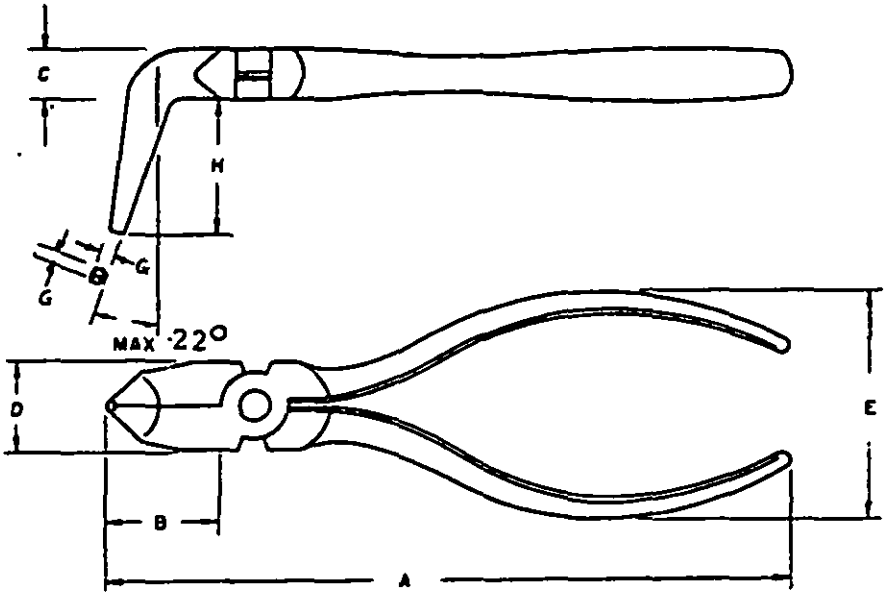


Figure 21. Type XII, class 1

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3.19.2 Class 2.

3.19.2.1 Style A. The jaws of style A pliers shall be straight and shall taper from near the joint section to the outermost end. The pliers shall conform to the requirements shown in table 20, and shall be similar to figure 22.

TABLE XX. Type XII, class 2, styles A and B

Nominal size	A ± 1/2	B ± 5/16	C ± 1/16	D + 3/16 - 1/8	E + 1/2 - 1/4	F ± 1/16	G ± 1/16	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inch	Inches	Inch	Inch	Inches
6-1/2	6-1/2	2-3/8	11/32	11/16	1-3/4	1/8	5/64	1-1/2

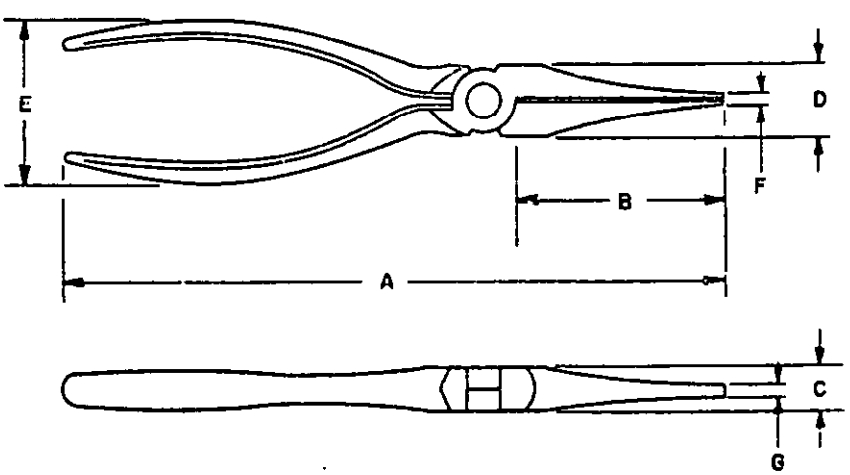


Figure 22. Type XII, class 2, styles A and B

3.19.2.2 Style B. Style B pliers shall conform to the requirements of 3.19.2.1, except that jaws shall be provided with cutting edges on one side adjacent to the joint and extending approximately one-fourth the length of the jaws. There shall be a recess behind the cutter to provide satisfactory clearance for the cutting edges.

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3.20 Type XIV.

3.20.1 Class 1. Class 1 pliers shall be of compound-leverage construction with parallel action jaws and side cutter. The pliers shall have an open throat and a longitudinal groove in one of the jaws suitable for gripping wire. Gripping surfaces of the jaws shall be suitably knurled to give a positive gripping surface. The longitudinal groove shall not be scored and shall be approximately 0.040 inch deep. With the pliers in the closed position, the jaw gripping surfaces shall firmly grip 0.016 inch diameter wire. The cutting edges of the cutters shall contact each other throughout their entire length in the closed position. Pliers shall conform to the requirements shown in table XXI and be similar to figure 23.

TABLE XXI. Type XIV, class 1

Nominal size	A ± 1/2	B ± 1/4	C ± 1/32	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inch
6-1/2 ---	6-1/2	2-3/16	3/8	17/32

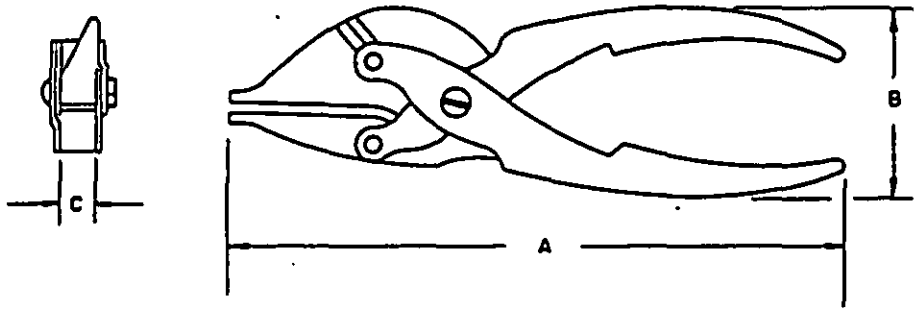


Figure 23. Type XIV, class 1

3.20.2 Class 2. Class 2 pliers shall be of compound-leverage construction with parallel action jaws. The pliers shall have an open throat and a longitudinal groove in one of the jaws suitable for gripping wire. Gripping surfaces of the jaws shall be suitably knurled to give a positive gripping surface and shall touch at the outermost end when the pliers are in a closed position. The longitudinal groove shall not be scored and shall be approximately 0.040 inch deep. Pliers shall conform to the requirements shown in table XXII and be similar to figure 24.

TABLE XXII. Type XIV, class 2

Nominal size	A ± 1/2	B ± 1/4	C - 1/32	Jaw opening, minimum
Inches	Inches	Inches	Inch	Inch
5 -------	5	1-7/8	1/4	3/8

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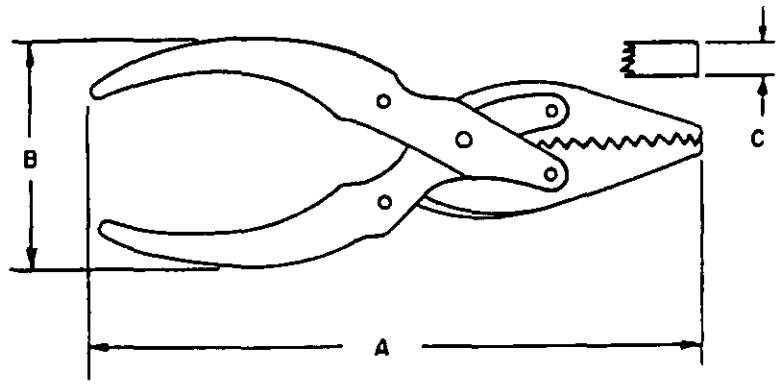


Figure 24. Type XIV, class 2

3.21 Type XV.

3.21.1 Class 1. The gripping jaws shall be of uniform width and shall taper in thickness from near the joint to the outermost end. The jaws shall contact each other at the outermost end when the pliers are in a closed position. Unless otherwise specified (see 6.1), the jaw-gripping surfaces shall be smooth and even and shall be without scoring. The pliers shall conform to the requirements shown in table XXIII and shall be similar to figure 25.

TABLE XXIII. Type XV, classes 1 and 2

Class	Nominal size	A ± 1/2	B ± 1/4	C ± 1/8	D ± 1/8	E ± 1/4	F ± 1/32	G ± 1/32	Jaw opening, minimum
	Inches	Inches	Inches	Inch	Inch	Inches	Inch	Inches	Inches
1 -----	6	6	2-3/16	5/16	11/16	2	3/32	---	1-1/2
2 -----	8	8	1-3/8	7/16	9/16	1-7/8	1/8	1/4	3/4

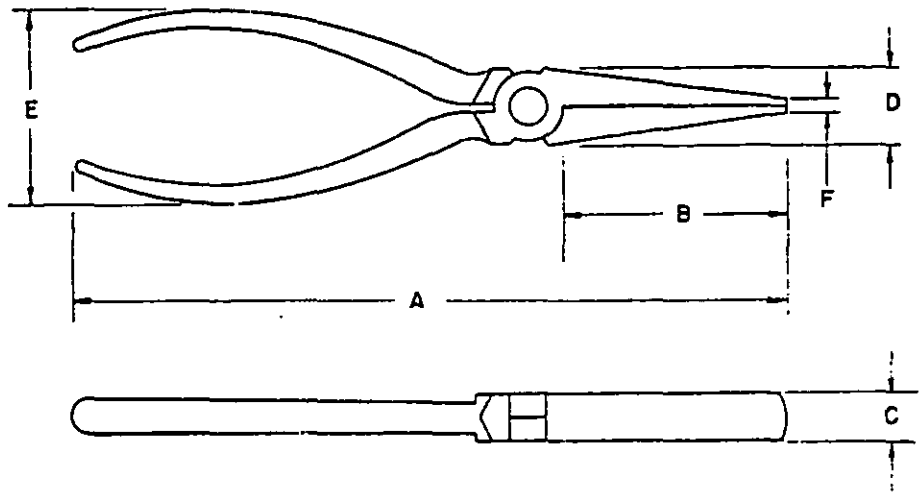


Figure 25. Type XV, class 1

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3.21.2 Class 2. The jaws of class 2 pliers shall taper uniformly in thickness and width from near the joint to the outermost end. Unless otherwise specified (see 6.1), the gripping surfaces of the jaws shall be scored. The length of the scoring shall be at least 1/2 inch back from the outermost end. When the pliers are in a closed position, the jaws shall contact each other at the outermost end. The pliers shall conform to the requirements shown in table XXIII and shall be similar to figure 26.

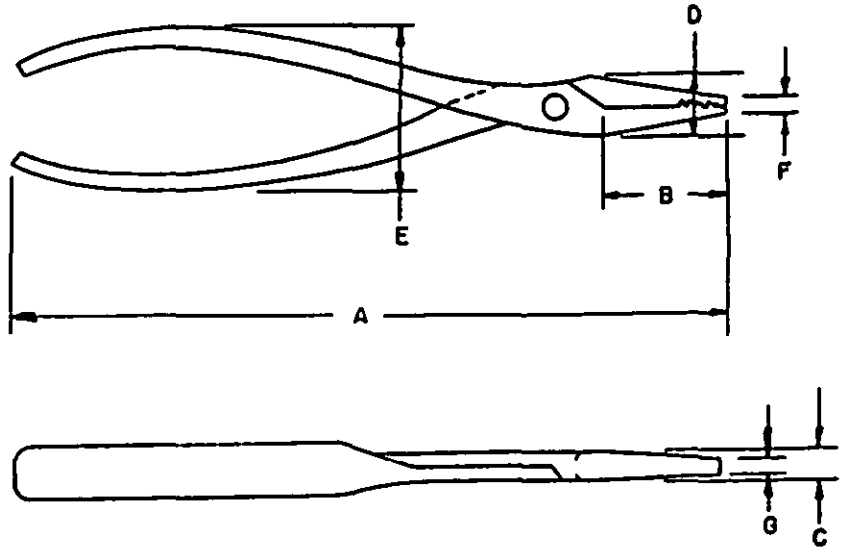


Figure 26. Type XV, class 2

3.22 Type XVI. Type XVI pliers shall be suitable for maintaining and installing wire fence on metal or wood posts. The head shall have a hammer on one jaw and a starting and pulling point on the other. Pliers shall be provided with two side wire cutters, and two eyes in the end of the head for twisting and splicing wire. The handles shall be provided with a milled grip adjacent to the joint suitable for wire stretching. Pliers shall conform to the requirements shown in table XXIV and shall be similar to figure 27.

TABLE XXIV. Type XVI

Nominal size	A ± 1/2	B ± 1/8	C ± 1/16	D ± 1/4	E ± 1/4
Inches	Inches	Inch	Inch	Inches	Inches
10 ------	10-3/8	3/4	9/16	3	2

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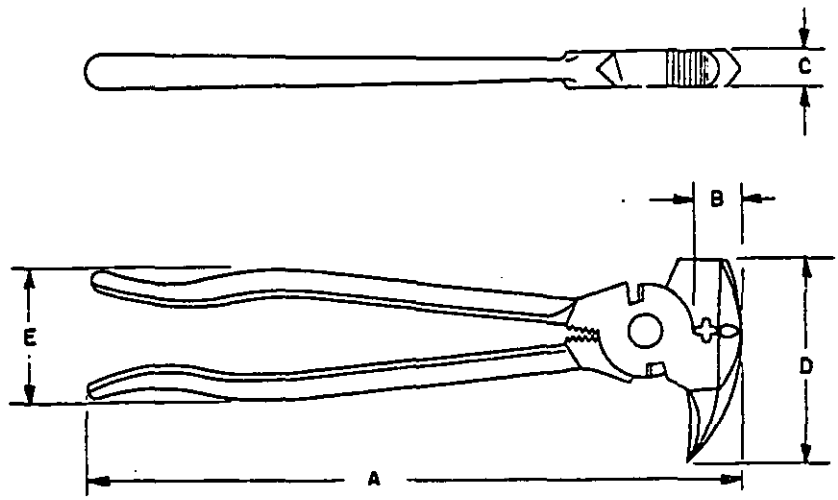


Figure 27. Type XVI

3.23 Type XVIII. The type XVIII pliers shall be designed primarily for removing heat coils from switchboards and telephone terminals. The gripping jaws shall be of uniform width and shall taper in thickness from near the joint to the outermost end. The nose at the outermost end shall be shaped to fit the coils as well as holding cylindrical objects. The jaw gripping surfaces shall be smooth and shall be without scoring. The pliers shall be capable of withstanding the test specified in 4.3.2. The pliers shall conform to the requirements shown in table XXV and shall be similar to figure 28.

TABLE XXV. Type XVIII

Nominal size	A ± 1/4	B ± 1/16	C ± 1/16	E ± 1/4	Transverse grip capacity	
					Minimum	Maximum
Inches	Inches	Inches	Inch	Inches	Inch	Inch
6 ------	6	1-3/16	3/8	2	.140	.312

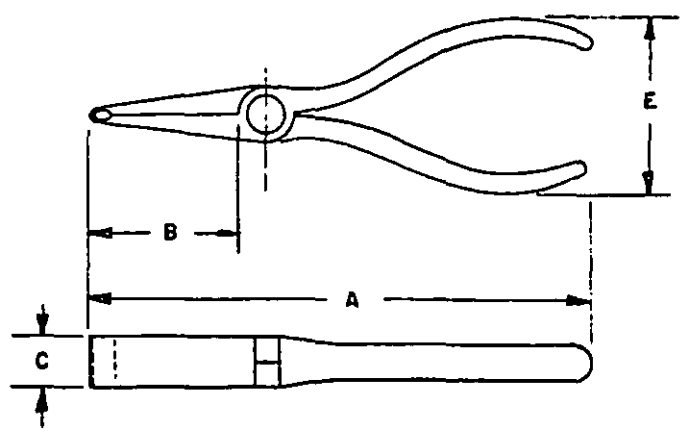


Figure 28. Type XVIII

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3.24 Workmanship. The workmanship on pliers covered by this specification shall be consistent with good industry practice. All surfaces shall be free from pits, rust, nodules, burrs, cracks, or other defects which may adversely affect the performance or appearance of the tool. The joint rivet (or bolt and nut) shall be properly tensioned to provide a snug fit of the two plier halves which will permit one-hand use, and yet is not so loose that the jaws will not be properly aligned.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with FED-STD-358, except as specified in 4.3.6.

4.3 Testing. Each sample unit shall be tested in accordance with 4.3.1 through 4.3.6 as applicable.

4.3.1 Hardness. Hardness tests shall be conducted in accordance with ASTM E-18. When grinding is necessary to prepare the test surface, the amount of material removed must not exceed 0.007 inches on the surface contacted by the indentor. Hardness determinations for handles, except cold formed steel handles (see 3.2), shall be taken approximately midway between the rivet and the end of the handles. With the exception of pliers with slip joints, the hardness of rivets or bolts shall be determined on the assembled tool. The hardness of bolts for type II pliers shall be taken on the flat faces of the cylindrical portion of the bolt (see 3.5).

4.3.1.1 Pliers with cutting edges. Hardness of the cutting edges shall be determined by the use of one test file having a known Rockwell "C" hardness of 52/54 and a second test file having a known Rockwell "C" hardness of 66/68. Each plier to be tested shall be mounted in a suitable clamping device and the test files applied to the cutting edges with short firm strokes. If the 52/54 file bites into the surface or the 66/68 file does not bite into the surface, the plier shall be rejected.

4.3.2 Handle-load tests. The sample plier under test shall be supported in a manner suitable for applying the load properly and determining the deflection and permanent set of handles. A platform scale similar to figure 29 may be used for making load tests. The gripping surfaces or cutting edges of the jaws shall be in normal horizontal position, with the head resting on a support or guide. The lower handle shall rest on a suitable fixed knife-edge support at the point of maximum handle curvature (the center of handle bow). Loads shall be applied through a knife edge to the upper handle at the point of maximum handle curvature (the center of handle bow) and in line with the lower handle support. Test loads applied to the handle shall be calculated by dividing the test load specified herein by the distance between the center of the rivet and the line of action of the load. Unless otherwise specified herein, steel inserts specified in connection with tests shall have a Rockwell C number of not less than 40. The width of the flat steel inserts shall be at least equal to that of the jaws. The method of determining the amount of

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deflection under load and the permanent set shall be that of marking the extreme ends of the handles with light centerpunch marks and measuring the distance between the marks by means of a divider and scale. After an initial application of a load of 5 pounds, the distance between the punch marks shall be measured with the load still applied. The difference between the initial measurement and the measurement after the test load has been reduced to 5 pounds, shall be the permanent set. If necessary, the extreme ends of the handles may be ground to obtain a surface suitable for making the centerpunch marks. Load and deflection tests may be made on a universal testing machine, a suitably equipped platform scale or a special calibrated testing device employing dead weights. If the deflection under load or the permanent set is in excess of the maximum value specified herein, or if the handles or jaws break, crack, chip, or spread under load, or if the rivet shows signs of weakness, the sample shall be rejected.

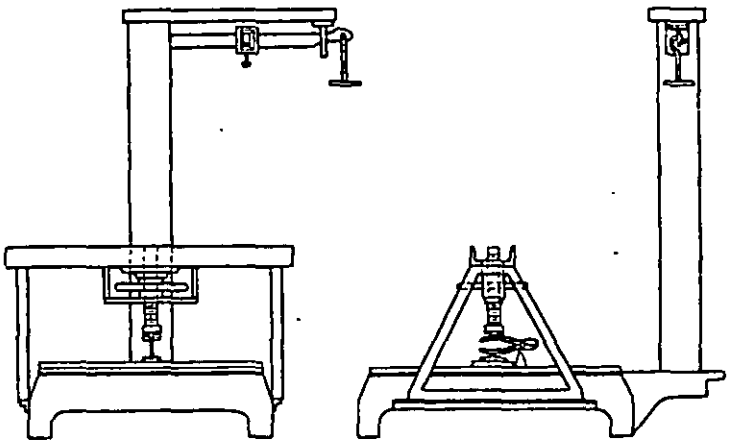


Figure 29. Apparatus for applying test load to handles of pliers

4.3.2.1 Allowable deflection and set. Except for types VI and XVIII pliers (see 4.3.2.1.1) the jaws of the pliers at their outermost end shall grip approximately 1/8 inch of a flat steel insert of the thickness shown in table XXVII, placed between the jaws. The deflection under load and permanent set of the handles shall not exceed the applicable values shown in table XXVII.

4.3.2.1.1 Allowable deflection and set for types VI and XVIII. With a hardened steel rod of the diameter shown in table XXVII, inserted between jaws of the large transverse pipe grip and the load applied at the point of maximum handle curvature (the center of handle bow), the deflection under load and permanent set of the handles shall not exceed the values shown in table XXVII.

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TABLE XXVII. Allowable deflection and permanent set for pliers

Type	Class and style	Nominal size	Test load	Deflection under load, maximum	Permanent set, maximum	Thickness of insert between jaws, (approximate)
		Inches	Inch - Pounds	Inch	Inch	Inch
I		7	200	0.53	0.13	1/2
II	1, style A	5	250	.50	.03	1/2
		8	450	.50	.03	1/2
		10 & 10A	750	.50	.03	1/2
		12	850	.50	.03	1/2
		16	950	.50	.03	1/2
II	2 and 3	6	675	.32	.03	1/8
		8	1000	.56	.06	1/8
		10	1450	.63	.13	1/8
III		8	1275	.65	.06	1/8
V	1	5-1/2	100	.50	.03	1/16
	2	4	100	.38	.03	1/16
	3	4	100	.38	.03	1/8
		6	480	.88	.03	1/8
VI		10	1400	.69	.06	1 [1]
VII		8	750	.50	.13	7/16
IX		6	600	.38	.03	1/8
		8	1020	.56	.06	1/8
X		4-1/2	100	.38	.03	1/16
		6	450	.88	.03	1/8
XI	1, style A	6	495	.94	.03	1/8
		7	495	.94	.03	1/8
XI	1, style B	4-1/2	100	.50	.03	1/16
		6	495	.94	.03	1/8
XI	2, style A	5	265	.45	.03	1/16
		6	495	.94	.03	1/8
XI	2, style B	4	100	.38	.03	1/16
		5	265	.45	.03	1/8
		6	400	.38	.03	1/8
		8	300	1.00	.03	1/8
XII	1	6	525	1.06	.03	1/16
XII	2, styles A and B	6-1/2	495	.94	.03	1/16
XIV	1	6-1/2	275	.250	.080	1/8
XV	1	6	480	.88	.03	1/8
	2	8	480	.88	.06	1/8
XVI		10	1275	.56	.06	1/8
XVIII		6	480	.88	.03	3/16 [1]

[1] Diameter of steel rod rather than thickness of flat insert.

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4.3.3 Cutting tests for pliers.

4.3.3.1 Steel wire. The wire for steel wire-cutting tests shall be uncoated single-strand improved plow steel wire having a tensile strength of 240,000 - 280,000 pounds per square inch (p.s.i.). This wire is employed for the plow steel main wires of wire rope. Where maximum loads are specified for wire cutting tests, equipment and manner of applying loads similar to that specified in 4.3.2 shall be used. The wire being cut may be loosely supported at the jaws by the operator in the manner ordinarily employed for cutting wire with pliers.

4.3.3.1.1 Steel-wire-cutting test. The pliers shall cut uncoated single-strand improved plow steel wire of the diameter shown in table XXVIII for the type, class and style plier specified (see 6.1). The cuts shall be made at the joint end of the cutting edges and the load required to completely sever the wire shall not exceed the applicable values shown in table XXVIII. Except for types III and XVI pliers, following completion of the cuts of wire, the pliers shall cut bond paper with the maximum load on the handles not exceeding that shown in table XXX (see 4.3.3.4).

TABLE XXVIII. Steel wire cutting test

Type	Class	Style	Nominal size	Wire diameter ± 0.002	Number of cuts	Handle load maximum
			Inches	Inch		Inch - Pounds
III	--	--	8	0.080	9	500
V	3	B	6	.062	3	480
IX	--	--	6	.080	3	600
	--	--	8	.091	3	1020
XI	1	A	6	.062	3	300
			7	.062	3	300
XI	2	A	5	.062	3	300
XII	2	B	6-1/2	.062	3	495
XIV	1	--	6-1/2	.080	3	275
XVI	--	--	10	.091	6	500

4.3.3.2 Lamp cord. Lamp cord shall be number 14 (0.0641 inch) American Wire Gage size, consisting of a twisted pair of conductors, each conductor comprising a cotton outer braid, rubber insulation, cotton separator and 41 copper-wire strands, each 10.0 mils in diameter. This wire is commercial type C lamp cord, number 14 (0.0641 inch) American Wire Gage, with cotton separator, conforming to the National Electric Code. The pliers under test shall cleanly cut the lamp cord.

4.3.3.2.1 Type IX. Type IX pliers shall cut lamp cord cleanly at any point along the cutting edges.

4.3.3.3 Nail-cutting test for type II, classes 2 and 3. Nails for nail-cutting tests shall be common steel wire nails. Three cuts shall be made manually. The cutter shall make three cuts of common steel wire nails of the sizes shown in table XXIX.

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TABLE XXIX. Nail sizes for nail-cutting test for type II, classes 2 and 3 pliers

Nominal size	Nail	
	Size	Diameter
Inches		Inch
6	6D	.113
8	8D	.131
10	10D	.148

4.3.3.4 Paper-cutting test. Paper for cutting tests shall be chemical wood, white bond paper conforming to type IV of Federal Specification UU-P-121. The bond paper shall be in strip form, the width of the strip measuring 1/16 inch less than the length of the cutter edges. The strip of paper shall be centered between the ends of the cutting edges when the cut is made. Equipment and manner of applying load similar to that specified in 4.3.2 shall be used. The paper cutting tests are intended to determine proper alignment of the cutting edges of the wire cutter. Paper-cutting tests which follow wire-cutting tests are intended to determine whether deformation or failure has occurred due to crushing of the cutting edges of the wire cutter incident to cutting wire. The cutters of the pliers under test shall completely and cleanly cut the bond paper when subjected to the test load specified in table XXX.

TABLE XXX. Paper cutting test

Type	Class	Style	Nominal size	Test load maximum
			Inches	Pounds
V	3	B	6	75
IX	-	-	All sizes	75
XI	1	A	All sizes	75
XI	2	A	All sizes	75
XII	2	B	6-1/2	75
XIV	1	-	All sizes	75

4.3.4 Wire-bending tests. The wire for the bending test shall be uncoated single-strand improved plow-steel wire having a minimum tensile strength of not less than 240,000 - 280,000 p.s.i. The test wire shall be held in a bench vise with approximately 4 inches of its length extending vertically beyond the vise jaws. Approximately 1 inch above the top of the vise jaws, the wire shall be securely gripped 3/8 inch from the nose end of type XII, class 2 long nose plier jaws; and 1/8 inch from the end of type V, classes 2 and 3, types X, XI, XIV, and type XV plier jaws. Bend the wire 90 degrees, then reverse bend through 180 degrees and repeat until the number of bends specified in 4.3.4.1 are made. Each 90-degree movement of the wire shall be counted as one bend. The radius of the bend shall be the minimum practicable when the most secure grip on the wire is obtained manually.

4.3.4.1 The type V, classes 2 and 3, type X, type XI, type XII, class 2, type XIV, and type XV pliers shall make a total of six 90-degree bends of uncoated, single strand, improved plow-steel wire of the diameter shown in table XXXI. If wire breaks during the testing, the wire shall be regripped and bend until the six complete 90° bends have been made.

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TABLE XXXI. Wire sizes for wire bending tests

Type	Class	Style	Nominal size	Wire diameter ± 0.002
			Inches	Inch
V	2	--	4	0.034
V	3	A	4	.034
		B	6	.062
X	--	--	4-1/2	.047
	--	--	6	.062
XI	1	A	6	.047
			7	.047
XI	1	B	4-1/2	.034
			6	.047
XI	2	A	5	.062
XI	2	B	4	.062
			5	.062
			6	.062
			8	.062
XII	2	A and B	6-1/2	.062
XIV	1	--	6-1/2	.062
XIV	2	--	5	.034
XV	1	--	6	.06
	2	--	8	.062

4.3.5 Jaw-opening test. One handle of the sample plier under test shall be clamped in a vise, with the pliers in a horizontal position, and a force shall be applied to the unclamped handle in such a direction as to tend to open the jaws of the pliers. The force shall be applied in the plane of the handles by means of a spring scale or other suitable load indicating device, at a distance from the center of the joint rivet equal to the moment arm distance specified herein for the handle load test or, if this distance is not specified, at the point of maximum handle curvature. The force required to open the jaws of the pliers to the respective minimum distance specified for the individual types, styles, and sizes shall be not greater than that shown in table XXXII.

TABLE XXXII. Jaw opening test

Nominal size	Force applied to handle, maximum
Inches	Pounds
4 and 4-1/2	1
5 and 5-1/2	2
6 and over	3

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